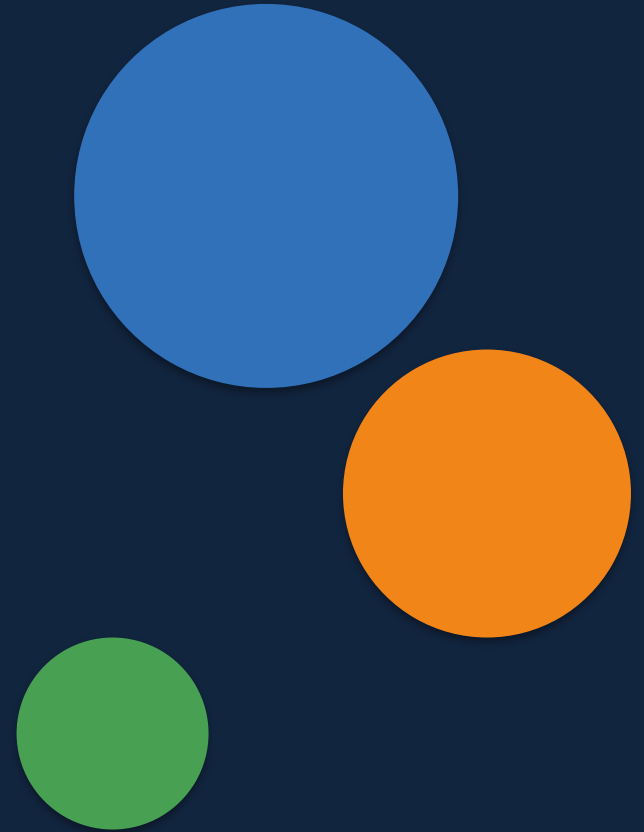


ChurnGuard AI

A data-driven early warning and retention system for Company A

Final Assignment | GCI World 2026 April



Prepared for: Company A, Telecommunications Service Provider

Executive Summary

Company A has a high near-term churn challenge. The model does not replace business judgment; it prioritizes the customers who should receive retention actions first.

49.6%

Observed churn in 31-60 days

0.667

XGBoost ROC-AUC on holdout

75.6%

Actual churn in top risk decile

1.9M

Potential preserved billing units

Core problem

Churn is broad, but marketing budgets are limited. Company A needs to know who to contact first and what action to take.

Proposed solution

A monthly churn-risk scoring system combined with action segments: service quality, usage decline, device age and high-value risk.

Business value

Target the top 20% risk customers instead of giving generic discounts to the whole base.

Market Context: Retention Is a Profit Lever

The mobile market is large and increasingly data-intensive; network experience and personalization are central competitive factors.

As mobile data traffic grows, service quality issues become more visible to customers and more expensive for operators.

External evidence supports analytics-based retention: advanced analytics can reduce telecom churn by up to 15%.

Retaining customers is economically important because customer acquisition is often much more expensive than retention.

Why this matters for Company A

The dataset already contains behavior, revenue, service-quality and device variables. These can be converted into a retention decision system.

Strategic direction

Move from broad reactive offers to targeted interventions based on predicted churn risk and customer value.

PoC objective

Prove that existing customer data can prioritize retention actions before implementing a full production system.

Data and Business Objective

Dataset provided by Company A: 100,000 customer records across two tables.

Client.csv

Customer profile, household, handset, account and billing-history variables.

Record.csv

Monthly usage, calls, service quality, customer care, revenue and churn label.

Target: churn

Instance of churn between 31-60 days after the observation date.

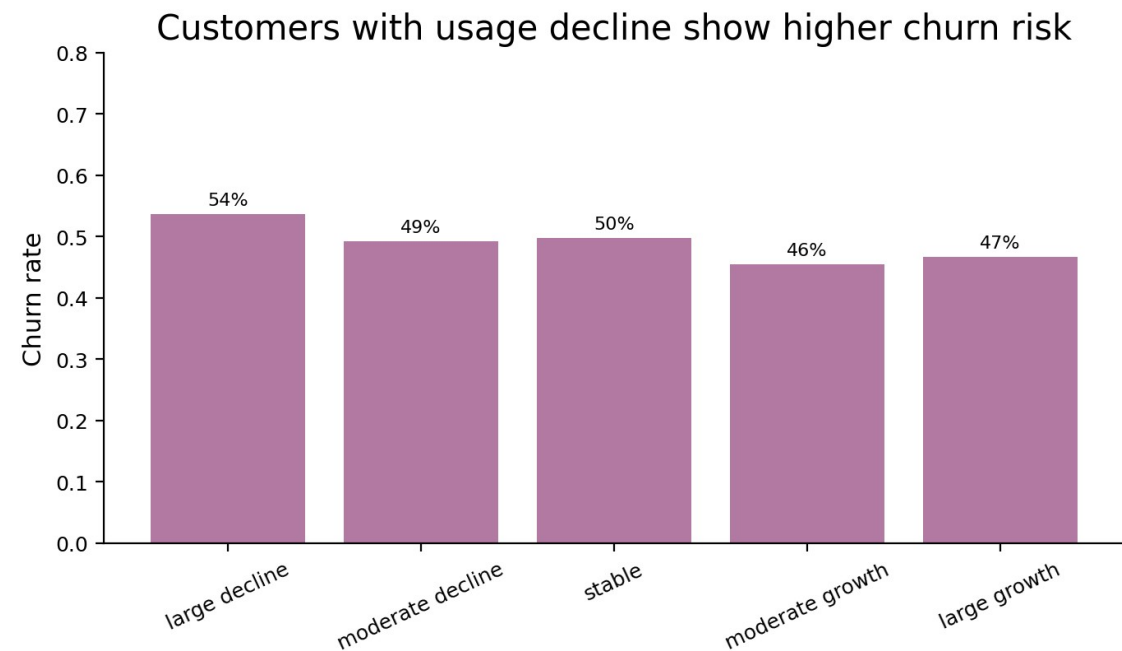
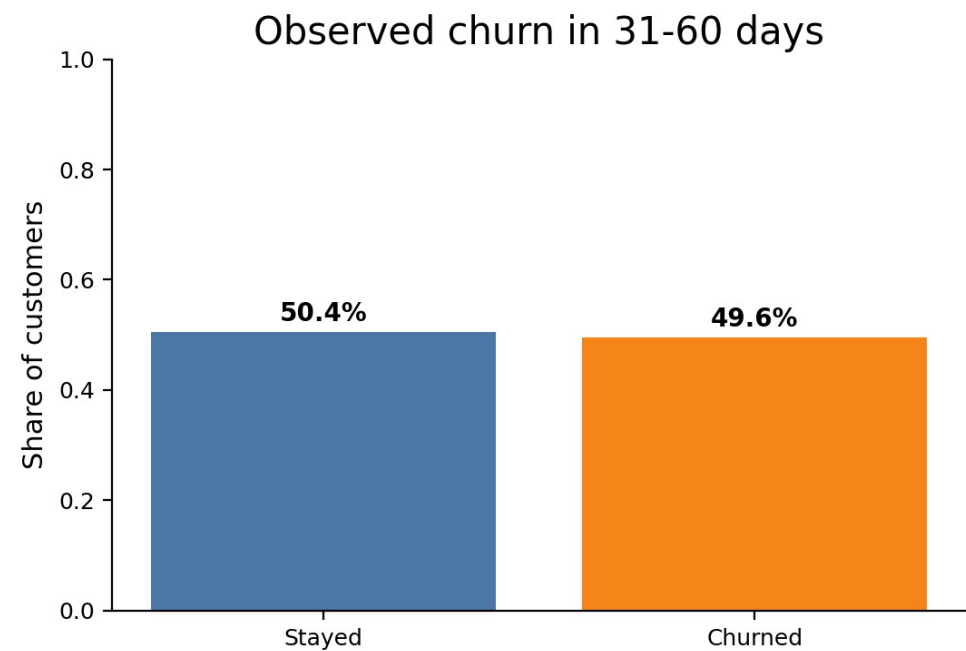
Business objective: reduce avoidable customer churn while protecting high-value revenue.

Machine learning task: binary classification to estimate churn probability.

Evaluation metric: ROC-AUC, because the business need is customer ranking and prioritization.

Output action: a monthly customer list for retention outreach and campaign design.

EDA: Churn Is a Large and Immediate Business Issue

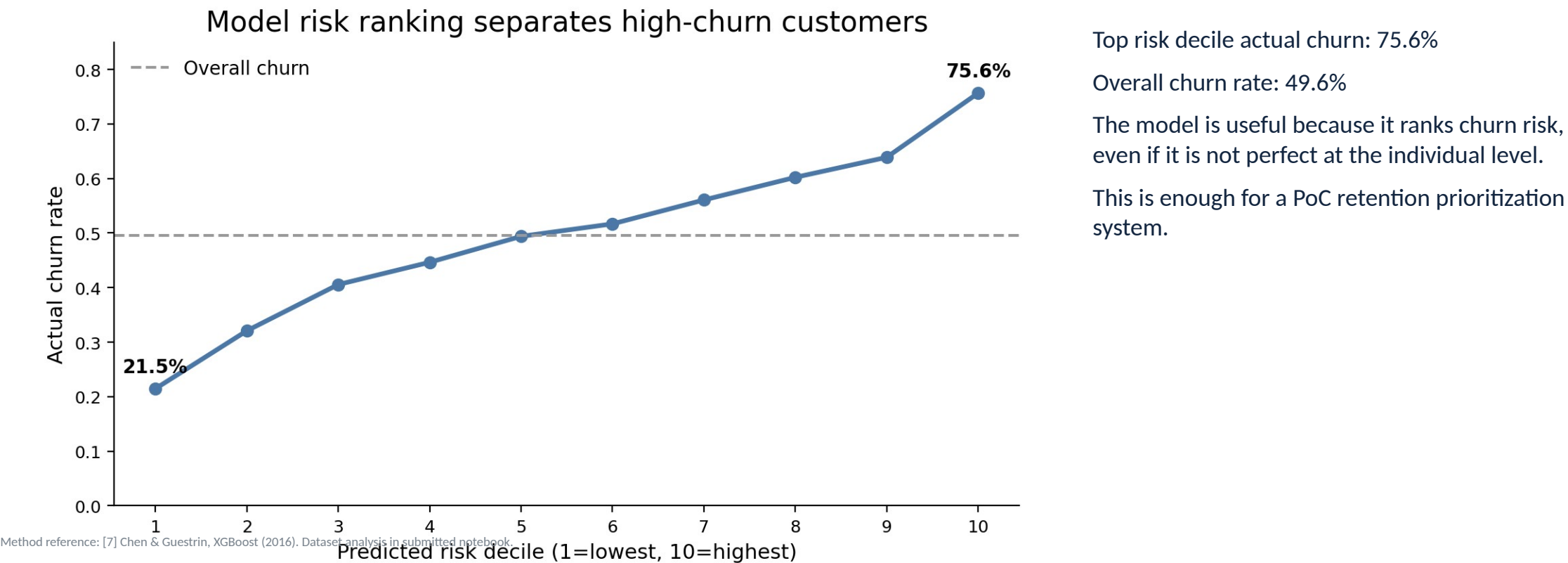


Key insight: nearly half of observed customers churned, and recent usage decline is a visible warning signal. This supports an early-warning system rather than only end-of-contract campaigns.

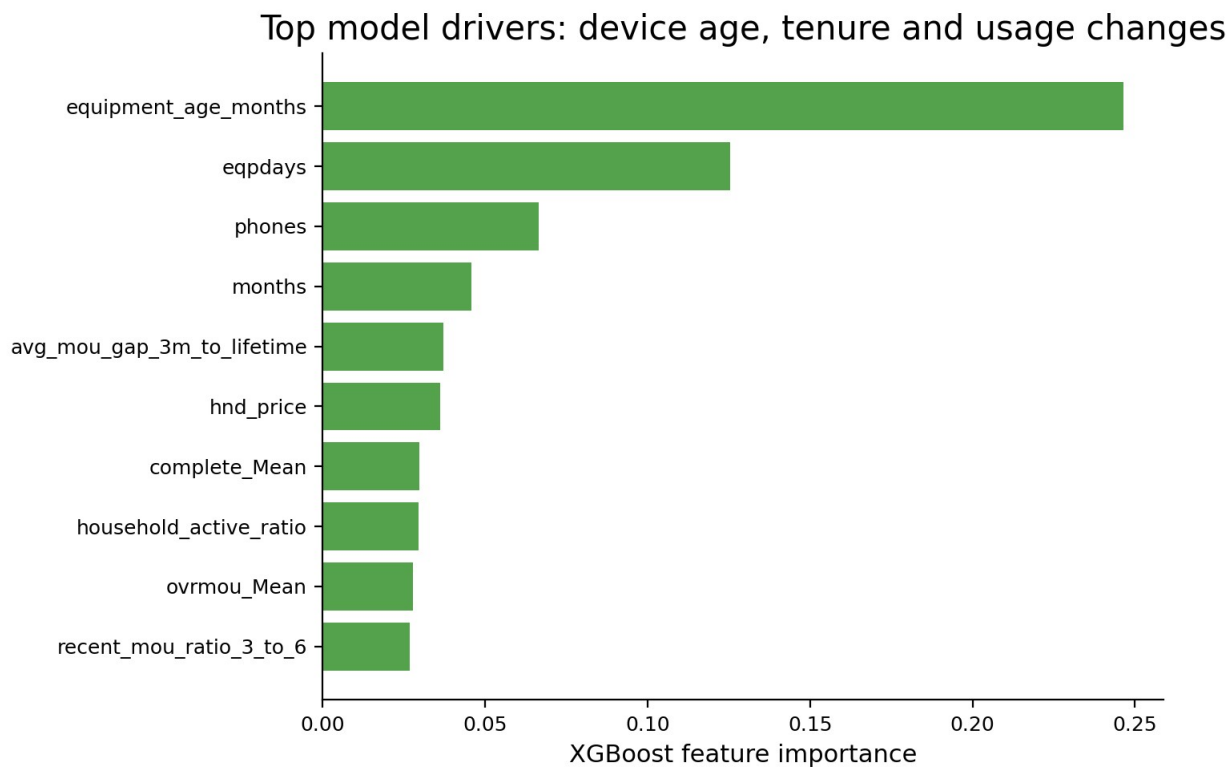
Source: Company A provided dataset; analysis in submitted notebook.

Model: Predicting Churn Risk for Prioritized Action

Model	Metric	Score
XGBoost Gradient-Boosted Decision Trees	ROC-AUC on stratified 20% holdout test set	AUC = 0.667; AP = 0.651



What Drives Churn Risk?



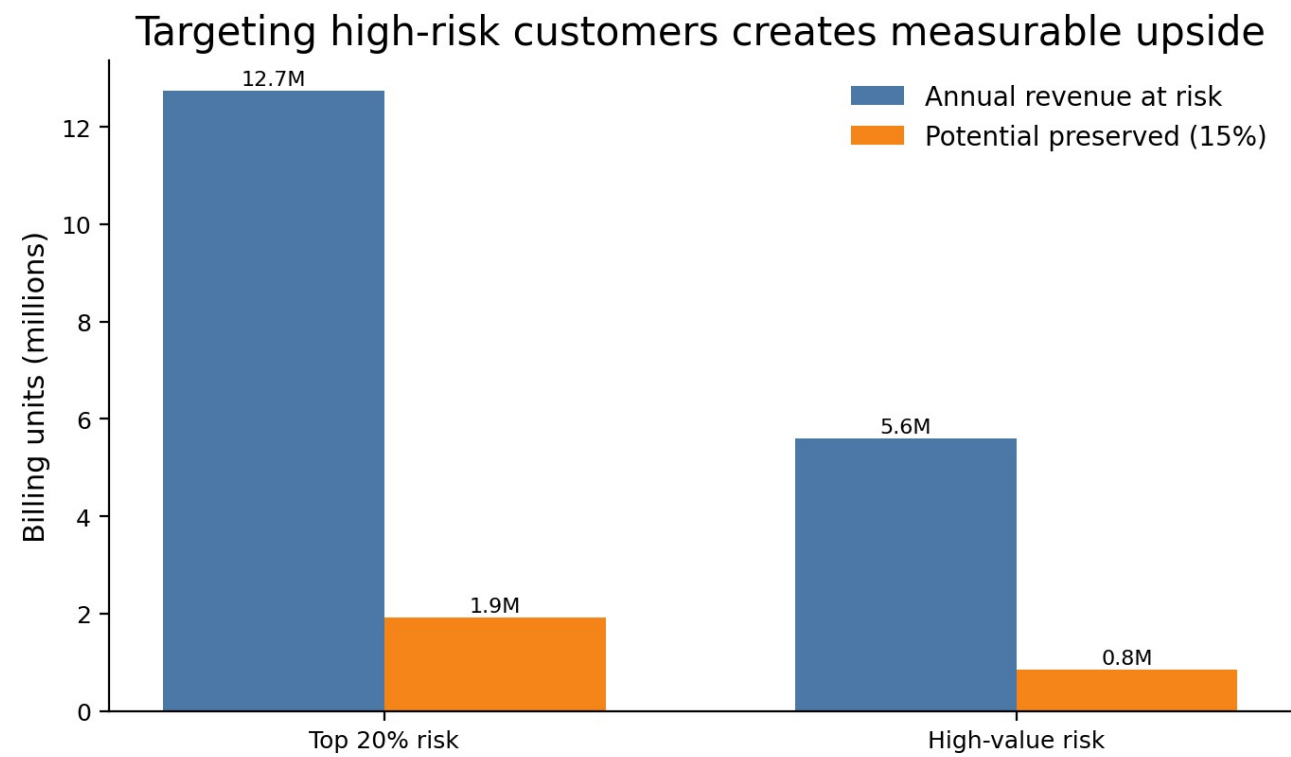
The strongest predictors are not only demographics. They are operational and behavioral:

- Equipment/device age and number of phones issued
- Customer tenure and handset price
- Recent usage decline vs lifetime average
- Overage behavior and plan fit
- Service interaction signals such as customer care and failed calls

Business implication: actions should be segmented. A customer with device-age risk needs a different offer from a customer with service-quality risk.

Source: Company A provided dataset; XGBoost feature importance from submitted notebook.

Priority Segment: High Risk Meets High Revenue



20,000
customers in top 20% predicted risk

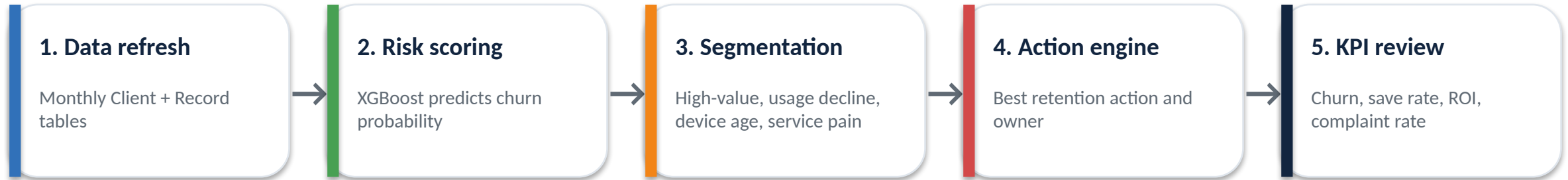
69.8%
actual churn rate in top 20% risk

5.6M
annualized high-value revenue at risk

Note: revenue is reported as Company A billing units because the original currency was not disclosed.

Source: Company A provided dataset; analysis in submitted notebook.

Proposal: ChurnGuard AI Operating Workflow



Operating principle

Use prediction as a prioritization tool, not as an automatic penalty.

Focus expensive human outreach on high-risk and high-value customers.

Run A/B tests to estimate real campaign lift before scaling.

From Prediction to Action: Four Retention Plays

High-value risk

Action: priority retention call + personalized loyalty offer. KPI: save rate and retained revenue.

Usage decline

Action: plan-fit review, usage-based bundle, engagement message. KPI: recovery in minutes/use.

Device-age risk

Action: handset upgrade or renewal offer. KPI: upgrade conversion and churn reduction.

Service pain

Action: network/customer-care follow-up. KPI: complaint resolution and drop/block reduction.

Why targeted actions matter

Generic discounts waste money on customers who would stay anyway.

A churn score plus business segmentation connects data science to operational decisions.

The campaign can start as a controlled PoC before full-scale automation.

Quantified Impact: Conservative PoC Case

Assumption for pilot sizing: analytics-driven retention reduces churn in the targeted population by 15%. This is aligned with external telecom analytics evidence and should be validated by A/B testing.

12.7M

Annualized revenue at risk in top 20% risk

1.9M

Potential preserved if churn falls 15%

0.84M

Preserved from high-value risk subgroup

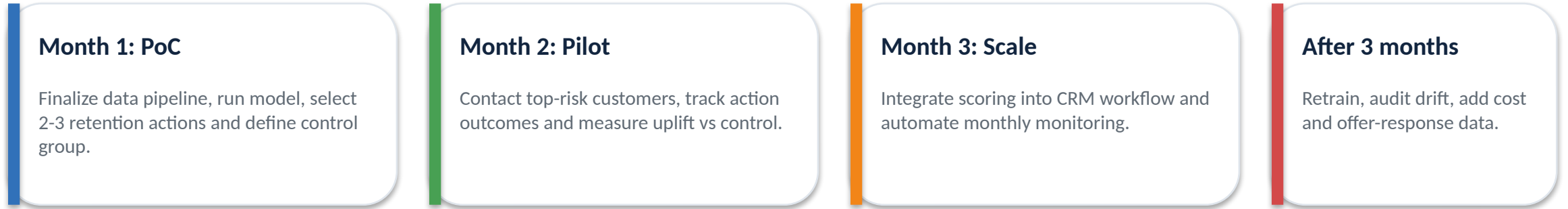
Primary KPI: churn rate reduction among contacted high-risk customers.

Financial KPI: retained monthly revenue minus offer and contact costs.

Operational KPI: customer-care resolution rate and device-upgrade conversion.

Decision gate: continue only if pilot ROI is positive and statistically credible.

Implementation Roadmap and Risk Control



Risk controls

Human approval for sensitive customer treatment; no automated termination decisions.

Monitor model drift because usage, devices and pricing can change over time.

Avoid unfair targeting by auditing outcomes across customer groups.

Measure uplift with controlled experiments before expanding budget.

Recommendation and References

Recommendation: approve a 3-month ChurnGuard AI PoC focused on the top 20% predicted-risk customers, with a priority lane for high-value accounts. Scale only after uplift and ROI are confirmed.

[1] GSMA. The Mobile Economy 2025. <https://www.gsma.com/solutions-and-impact/connectivity-for-good/mobile-economy/the-mobile-economy-2025/>

[2] Ericsson Mobility Report. Mobile data traffic forecast. <https://www.ericsson.com/en/reports-and-papers/mobility-report/dataforecasts/mobile-traffic-forecast>

[3] McKinsey & Company. Reducing churn in telecom through advanced analytics. <https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/reducing-churn-in-telecom-through-advanced-analytics>

[4] Saleh et al. Customer retention and churn prediction in the telecommunication industry. PMC. <https://pmc.ncbi.nlm.nih.gov/articles/PMC10239051/>

[5] GCI World 2026 April. Company A Dataset Overview. Provided course material.

[6] GCI World 2026 April. Final Assignment README and tutorial. Provided course material.

[7] Chen, T. and Guestrin, C. XGBoost: A Scalable Tree Boosting System. <https://doi.org/10.1145/2939672.2939785>

[8] Generative AI support: ChatGPT conversation URL should be pasted in the Omnicampus reference field if required.